



Delivery of Suomi National Polar-orbiting Partnership (Suomi NPP) satellite data products will cease on November 1, 2026. Data users should transition now to alternative products from NOAA-21 and NOAA-20. Suomi NPP data announcement



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Earth Observation Data Basics

The life cycle of Earth observation data is rich and complex, with many points of entry along the pipeline. From collection to visualization, we dive deep into the basics to demystify the incredible data in our catalog.

Remote Sensing

Remote sensing is the acquiring of information from a distance. NASA observes Earth and other planetary bodies via remote instruments on space-based platforms (e.g., satellites or spacecraft) and on aircraft that detect and record reflected or emitted energy. Remote instruments, which provide a global perspective and a wealth of data about Earth systems, enable data-informed decision making based on the current and future state of our planet.

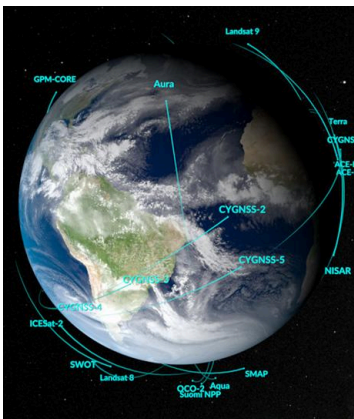
For more information, check out the [Fundamentals of Remote Sensing training](#) from the Applied Remote Sensing Training (ARSET) program.

Get Data

Explore the world's largest repository of open-access Earth science data.

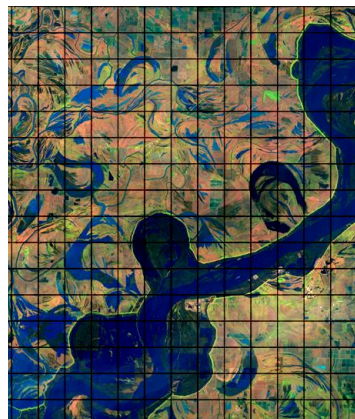
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Remote Sensing Data Basics



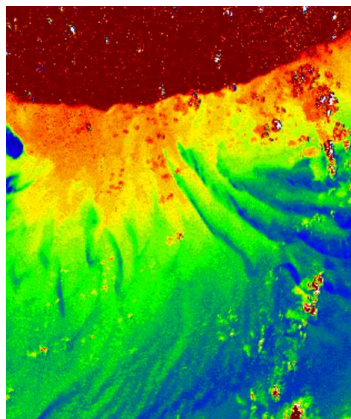
Orbits

An orbit is the curved path a satellite follows around the Earth due to



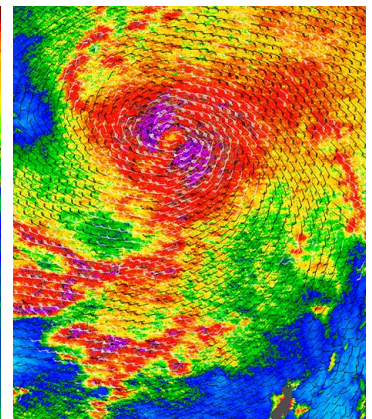
Resolution

There are four types of resolution to consider for any dataset — radiometric,



The Electromagnetic Spectrum

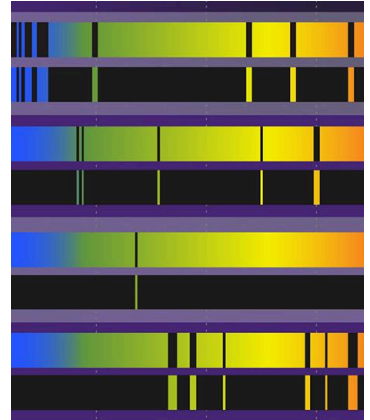
To understand remotely sensed data, it is imperative to know the electromagnetic



Active Instruments

Active instruments emit energy and collect data based on changes in the

return signal.



Field Spectroscopy

Spectroscopy is the measurement of how materials interact with electromagnetic radiation across the spectrum. Scientists have tools to do this from the ground and ships.

This interactive tool helps users navigate and understand essential metadata on our Earth science dataset landing pages. Through guided examples and hands-on exploration, learn critical context about our data to aid you in your own scientific discoveries.



Cloud Computing

Nearly all of NASA's Earth science data is accessible through Earthdata Cloud, making access, analysis, and visualization more efficient and cost effective. We offer resources including Python libraries, tutorials, and data recipes to help users optimize working with data in the cloud.

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Earth Observation Data and Artificial Intelligence

The application of artificial intelligence (AI) to Earth science data makes it possible to search through large amounts of data to find relationships.

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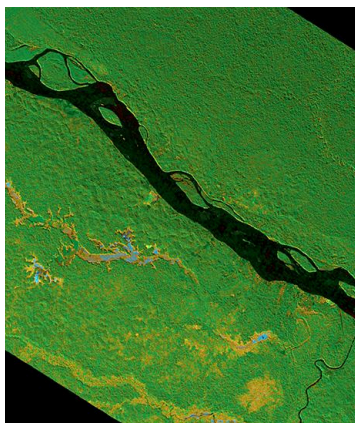
Synthetic Aperture Radar

Synthetic aperture radar (SAR) is a type of remote sensing that produces fine-resolution data using a technology that, over time, can detect even minute changes on Earth's surface.



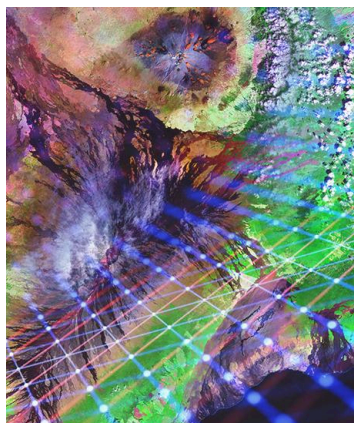
Synthetic Aperture Radar (SAR)

SAR is one of the power technologies of remote sensing, and enables high resolution imagery to be created night or day, regardless of weather conditions.



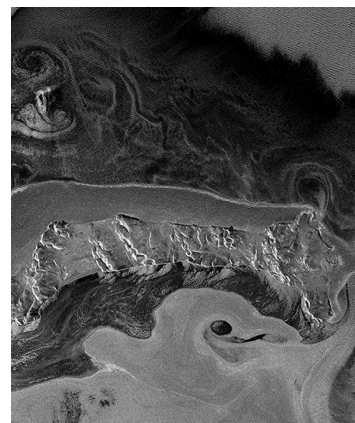
The SAR Handbook

The SAR Handbook was created in 2019 as a guide for forest monitoring and biomass estimation with synthetic aperture radar (SAR).



Types of SAR Products

View a table of synthetic aperture radar (SAR) products and their processing levels available through NASA's Earth Science Data Systems (ESDS) Program.



SAR Image Interpretation

General rules of thumb for interpreting synthetic aperture radar (SAR) imagery and resources for viewing SAR imagery.

Join Our Community of NASA Data Users

While NASA data are openly available without restriction, an Earthdata Login is required to download data and to use some tools with full functionality.

[Learn About the Benefits of Earthdata Login](#) 

QUESTIONS & SUPPORT

Have a question?

Get help with NASA Earth science data, datasets, tools, services, and metadata on the Earthdata Forum. Experienced staff in the data user community review and answer questions.



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Data Catalog

Dive into one of the planet's largest repositories of Earth science data.

Data Tools

Access a range of tools that enable data discovery, analysis, and visualization.



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